(chosen randomly), I found more than half (30 out of 58) were from third parties; that is, they were not from the website itself.

Token-based authentication is widely used. The OAuth 2.0 protocol, for example (which we'll discuss in detail in chapter 12), uses authenticated tokens. OAuth 2.0 is the basis of the OpenID Connect protocols that are used in social login systems that implement single-sign-on using a social networking account like those from Google, Facebook, and X. Token-based authentication is also often used in delegated authentication flows.

3.6.3 Identifier with authentication factors

Identifier with authentication factors is the most commonly used authentication method. Let's examine how authentication factors are used together to prove control over an identifier and its associated account.

When Alice logs in, she provides an identifier to signal which account she is claiming. She also presents one or more authentication factors to prove that the identifier belongs to her. These factors may be provided simultaneously along with the identifier or as part of a workflow, a process known as *progressive disclosure*.

Multi-factor authentication, where two or more factors are combined, is now quite common, although it is often optional. Passwords are usually combined with a possession factor in the form of a one-time password (OTP) to make it harder for an attacker to gain access to Alice's account using a stolen, guessed, or phished password. OTPs are possession factors because they depend on Alice having access to the means by which the OTP is either delivered to her or generated. OTPs can take the form of SMS or email messages with a PIN, a push notification on an app, or a code from an authentication app such as Authy or Microsoft Authenticator. Figure 3.7 shows the flow for authentication using an identifier and authentication factors.